Unit 15 Submission File: Web Vulnerabilities and Hardening
Part 1: Q&A
The URL Cruise Missile
The URL is the gateway to the web, providing the user with unrestricted access to all available online resources. In the wrong hands can be used as a weapon to launch attacks.
Use the graphic below to answer the following questions:

Protocol	**Host Name**	**Path**	**Parameters**
http://	www.buyitnow.tv	/add.asp	?item=price#1999

1. Which part of the URL can be manipulated by an attacker to exploit a vulnerable back-end database system?
Answer: Parameters
2. Which part of the URL can be manipulated by an attacker to cause a vulnerable web server to dump the `/etc/passwd` file? Also, name the attack used to exploit this vulnerability.
Answer: Path

3. Name three threat agents that can pose a risk to your organization.
Answer: Hackers, Government, Internal Employees
4. What kinds of sources can act as an attack vector for injection attacks?
Answer: Server directory, form, website, login page, application session
5. Injection attacks exploit which part of the CIA triad?
Answer: Confidentiality
6. Which two mitigation methods can be used to thwart injection attacks?

Answer: Input Sanitization & Validation

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Web Server Infrastructure
Web application infrastructure includes sub-components and external applications that provide efficiency, scalability, reliability, robustness, and most critically, security.
- The same advancements made in web applications that provide users these conveniences are the same components that criminal hackers use to exploit them. Prudent security administrators need to be aware of how to harden such systems.
Use the graphic below to answer the following questions:

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Client	Firewall	Webserver	Web Application	Database

1. What stage is the most inner part of the web architecture where data such as, customer names, addresses, account numbers, and credit card info, is stored?
Answer: Stage 5 SQL Database
2. Which stage includes online forms, word processors, shopping carts, video and photo editing, spreadsheets, file scanning, file conversion, and email programs such as Gmail, Yahoo and AOL.
Answer: Stage 4 Web Application
3. What stage is the component that stores files (e.g. HTML documents, images, CSS stylesheets, and JavaScript files) that's connected to the Internet and provides

support for physical data interactions between other devices connected to the web?
Answer: Stage 3 Web Server
4. What stage is where the end user interacts with the World Wide Web through the use of a web browser?
Answer: Stage 1 Web Client
5. Which stage is designed to prevent unauthorized access to and from protected web server resources?
Answer: Stage 2 Firewall

Server Side Attacks
In today's globally connected cyber community, network and OS level attacks are well defended through the proper deployment of technical security controls such as, firewalls, IDS, Data Loss Prevention, EndPoint and security. However, web servers are accessible from anywhere on the web, making them vulnerable to attack.
1. What is the process called that cleans and scrubs user input in order to prevent it from exploiting security holes by proactively modifying user input.
Answer: Input Sanitization
2. Name the process that tests user and application-supplied input. The process is designed to prevent malformed data from entering a data information system by verifying user input meets a specific set of criteria (i.e. a string that does not contain standalone single quotation marks).
Answer: Input Validation
3. **Secure SDLC** is the process of ensuring security is built into web applications throughout the entire software development life cycle. Name three reasons why organization might fail at producing secure web applications.
Answer: **High Implementation Costs**
Insufficient support from management & standardization
<u>No Quality management</u>
<u>Total reliance on web application firewalls</u>
<u>Reactive Security postures</u>
4. How might an attacker exploit the `robots.txt` file on a web server?
Answer: Reveal the existence information that a WebCrawler is hiding.
5. What steps can an organization take to obscure or obfuscate their contact information on domain registry web sites?
Answer: Ensure they have a service that protect privacy. Typically, a Registrar provides a service that perform the act of protecting an organizations privacy.

6. True or False: As a network defender, `Client-Side` validation is preferred over `Server-Side` validation because it's easier to defend against attacks.
- Explain why you chose the answer that you did.
Answer: False. Its harder to validate on the client side because the system/applications maintained at the client's discretion. Applications may have available exploits and provide malicious methods for attackers to utilize for impersonation.

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Web Application Firewalls

WAFs are designed to defend against different types of HTTP attacks and various query types such as SQLi and XSS.
WAFs are typically present on web sites that use strict transport security mechanisms such as online banking or e-commerce websites.
1. Which layer of the OSI model do WAFs operate at?
Answer: Layer 7 Application
2. A WAF helps protect web applications by filtering and monitoring what?
Answer: HTTP/S Traffic
3. True or False: A WAF based on the negative security model (Blacklisting) protects against known attacks, and a WAF based on the positive security model (Whitelisting) allows pre-approved traffic to pass.
Answer:True

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Authentication and Access Controls
Security enhancements designed to require users to present two or more pieces of evidence or credentials when logging into an account is called multi-factor authentication.
- Legislation and regulations such as The Payment Card Industry (PCI) Data Security Standard requires the use of MFAs for all network access to a Card Data Environment (CDE).
- Security administrators should have a comprehensive understanding of the basic underlying principles of how MFA works.
1. Define all four factors of multifactor authentication and give examples of each:
   - Factor 1: Standard login inputs (password, PIN, cognitive questions)
   - Factor 2: Physical keys (smartcard, hard token)
   - Factor 3: Biometrics (iris/retina scan, hand geometry)
   - Factor 4: Location (GPS detection, callback to a home phone number)

2. True or False: A password and pin is an example of 2-factor authentication.
Answer: True

3. True or False: A password and `google authenticator app` is an example of 2-factor authentication.
Answer: True

4. What is a constrained user interface?

Answer: Restricts what users can see and do based on their privileges
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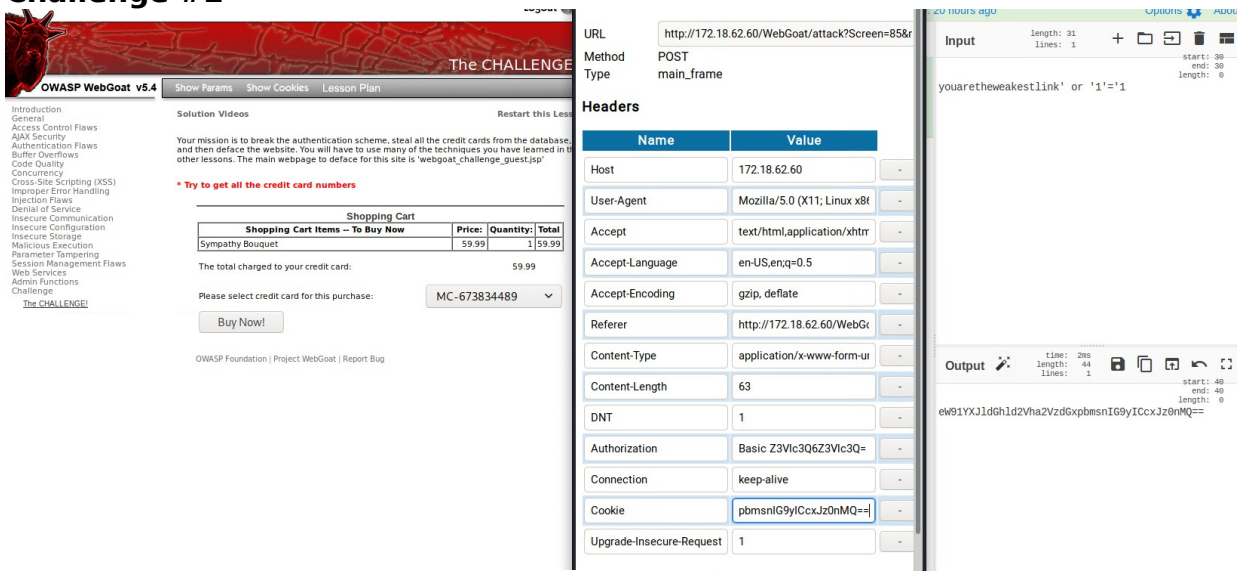
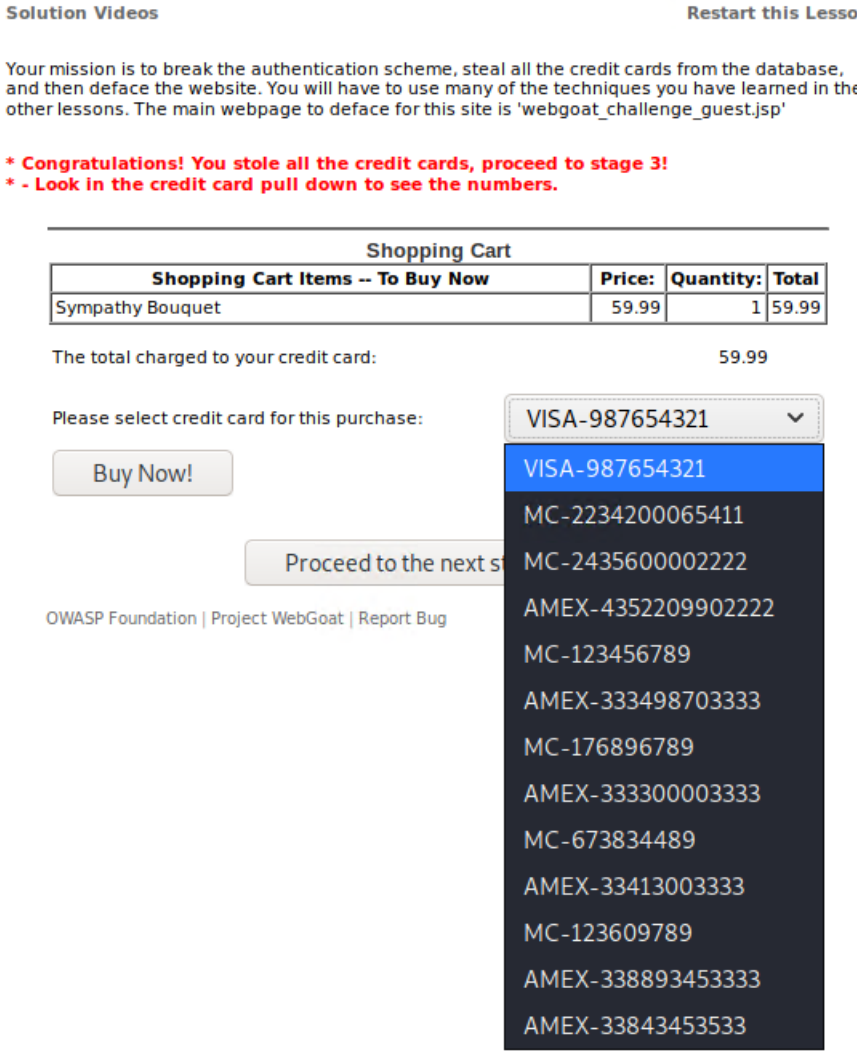
Part 2: The Challenge
The Challenge Instructions
Challenge #1

NetHunter Exploit-DB GHDB BeEF Authentication CyberChef

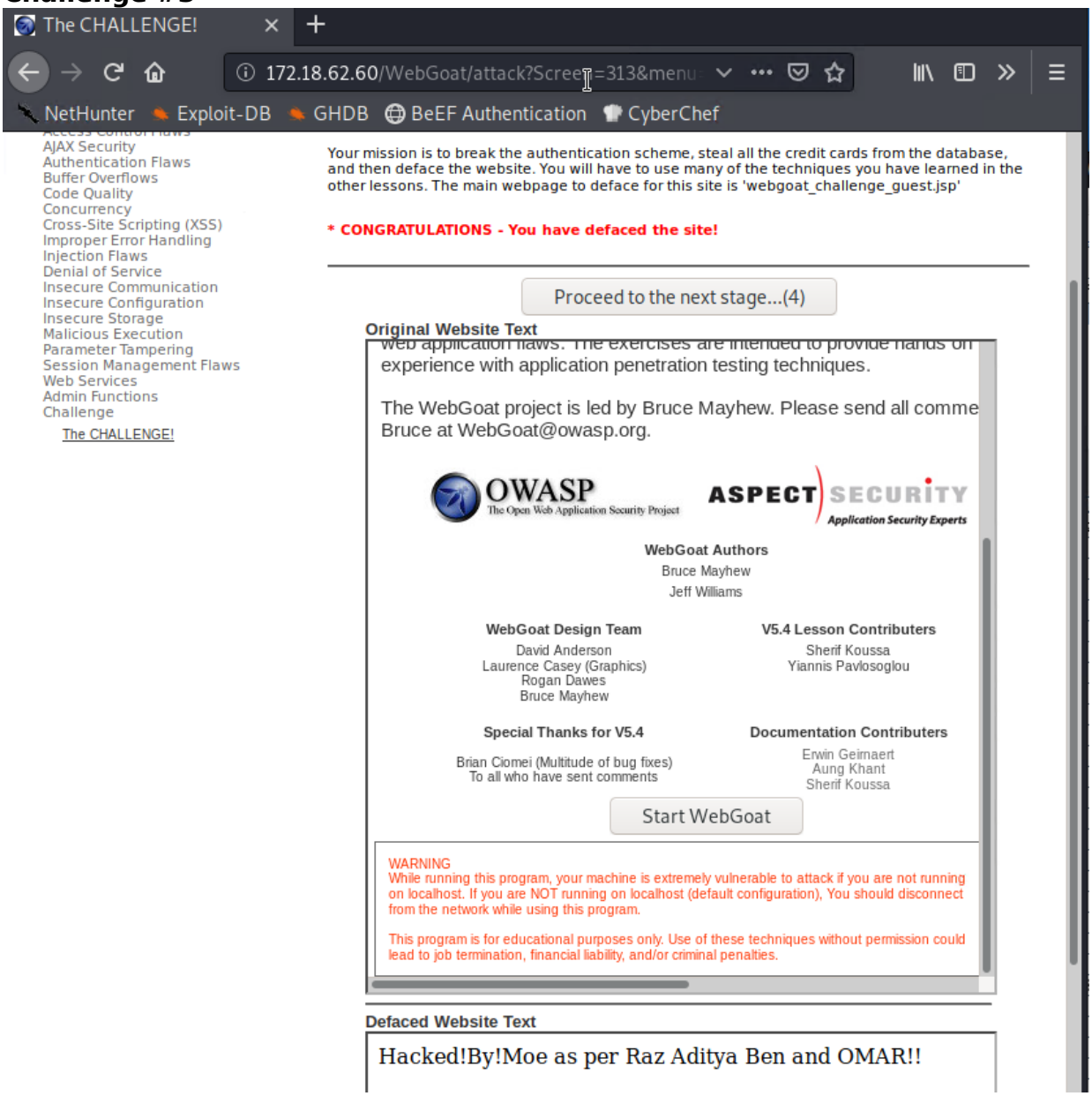
```
106     /**
107      * Description of the Field
108      */
109     protected final static String PASSWORD = "Password";
110
111     /**
112      * Description of the Field
113      */
114     protected final static String USER = "user";
115
116     /**
117      * Description of the Field
118      */
119     protected final static String USERNAME = "Username";
120
121     private String pass = "goodbye";
122
123     private String user = "youaretheweakestlink";
124
125     /**
126      * Description of the Method
127      *
128      * @param s
129      *          Description of the Parameter
130      * @return Description of the Return Value
131      */
132     protected Element createContent(WebSession s)
133     {
134         return super.createStagedContent(s);
135     }
136
137     /**
138      * Determine the username and password
139      *
140      * @param s
141      *          Description of the Parameter
142      * @return Description of the Return Value
```

Challenge #2


OWASP WebGoat v5.4
The CHALLENGE
Show Params
Show Cookies
Lesson Plan
Solution Videos
Restart this Lesson
Your mission is to break the authentication scheme, steal all the credit cards from the database, and then deface the website. You will have to use many of the techniques you have learned in the other lessons. The main webpage to deface for this site is 'webgoat_challenge_guest.jsp'
* Try to get all the credit card numbers
Shopping Cart
Shopping Cart Items -- To Buy Now
Price:
Quantity:
Total
Sympathy Bouquet
59.99
1
59.99
The total charged to your credit card:
59.99
Please select credit card for this purchase:
MC-673834489
Buy Now!
OWASP Foundation | Project WebGoat | Report Bug
URL
http://172.18.62.60/WebGoat/attack?Screen=85&
Method
POST
Type
main_frame
Headers
Name
Value
Host
172.18.62.60
User-Agent
Mozilla/5.0 (X11; Linux x86
Accept
text/html,application/xhtm
Accept-Language
en-US,en;q=0.5
Accept-Encoding
gzip, deflate
Referer
http://172.18.62.60/WebGo
Content-Type
application/x-www-form-ur
Content-Length
63
DNT
1
Authorization
Basic Z3Vlc3Q6Z3Vlc3Q=
Connection
keep-alive
Cookie
pbmsnIG9yICcxJz0nMQ==
Upgrade-Insecure-Request
1
Input
youaretheweakestlink' or '1'='1
Output
eW91YXJldGhld2Vha2VzdGxpbmsnIG9yICcxJz0nMQ==


Solution Videos
Restart this Lesso
Your mission is to break the authentication scheme, steal all the credit cards from the database, and then deface the website. You will have to use many of the techniques you have learned in the other lessons. The main webpage to deface for this site is 'webgoat_challenge_guest.jsp'
* Congratulations! You stole all the credit cards, proceed to stage 3!
* - Look in the credit card pull down to see the numbers.
Shopping Cart
Shopping Cart Items -- To Buy Now
Price:
Quantity:
Total
Sympathy Bouquet
59.99
1
59.99
The total charged to your credit card:
59.99
Please select credit card for this purchase:
VISA-987654321
Buy Now!
Proceed to the next st
OWASP Foundation | Project WebGoat | Report Bug
VISA-987654321
MC-2234200065411
MC-2435600002222
AMEX-4352209902222
MC-123456789
AMEX-333498703333
MC-176896789
AMEX-333300003333
MC-673834489
AMEX-33413003333
MC-123609789
AMEX-338893453333
AMEX-33843453533

Challenge #3


The CHALLENGE!
172.18.62.60/WebGoat/attack?Screen=313&menu
NetHunter
Exploit-DB
GHDB
BeEF Authentication
CyberChef
AJAX Security
Authentication Flaws
Buffer Overflows
Code Quality
Concurrency
Cross-Site Scripting (XSS)
Improper Error Handling
Injection Flaws
Denial of Service
Insecure Communication
Insecure Configuration
Insecure Storage
Malicious Execution
Parameter Tampering
Session Management Flaws
Web Services
Admin Functions
Challenge
The CHALLENGE!
Your mission is to break the authentication scheme, steal all the credit cards from the database, and then deface the website. You will have to use many of the techniques you have learned in the other lessons. The main webpage to deface for this site is 'webgoat_challenge_guest.jsp'
* CONGRATULATIONS - You have defaced the site!
Proceed to the next stage...(4)
Original Website Text
experience with application penetration testing techniques.
The WebGoat project is led by Bruce Mayhew. Please send all comme
Bruce at WebGoat@owasp.org.
OWASP
The Open Web Application Security Project
ASPECT SECURITY
Application Security Experts
WebGoat Authors
Bruce Mayhew
Jeff Williams
WebGoat Design Team
David Anderson
Laurence Casey (Graphics)
Rogan Dawes
Bruce Mayhew
V5.4 Lesson Contributers
Sherif Koussa
Yiannis Pavlosoglou
Special Thanks for V5.4
Brian Ciomei (Multitude of bug fixes)
To all who have sent comments
Documentation Contributers
Erwin Geirnaert
Aung Khant
Sherif Koussa
Start WebGoat
WARNING
While running this program, your machine is extremely vulnerable to attack if you are not running on localhost. If you are NOT running on localhost (default configuration), You should disconnect from the network while using this program.
This program is for educational purposes only. Use of these techniques without permission could lead to job termination, financial liability, and/or criminal penalties.
Defaced Website Text
Hacked!By!Moe as per Raz Aditya Ben and OMAR!!